



Terahertz immunosensing assisted by functionalized Au NPs based on all-dielectric metasurface

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ABSTRACT

Terahertz (THz) sensors have attracted significant attention in the biological field due to their non-destructive and harmless interaction with biochemical samples, but how to achieve efficiently specific sensing for the small molecular weight samples is still challenging. In this work, we incorporated an all-dielectric metasurface with the functionalized gold nanoparticles (Au NPs) to realize an immunosensing of the target biomolecule at nanomole-level concentration. These Au NPs are modified by the Anti-HA tag antibody, which realizes the specific immunobinding of the human influenza hemagglutinin tag protein (HA antigen). By comparing with and without the functionalized Au NPs, the experimental results show that this sensor can quantitatively detect the concentration of HA antigen, and its sensitivity can be greatly improved by the functionalized Au NPs. The maximum sensitivity can be achieved to 2.96 GHz•ml/nmol, which is 2.66 times higher than that without the Au NPs, and the highest detection accuracy reaches 1.05 nmol/ml. Furthermore, by comparing three other nonspecific proteins, only the HA antigen can be sensitively distinguished, which proves the specificity of this sensing method. Our results illustrate a reliable platform for THz immunosensing applications, and it is expected that the proposed sensing strategy opens new avenues for developing high-efficient, low-cost, and real-time THz all-dielectric sensors for biomolecular detection.

1. Introduction

Terahertz (THz) technology within an electromagnetic spectrum range of 0.1–10 THz (1 THz = 10^{12} Hz) has attracted great attention in biosensing [1]. Due to that THz waves are non-ionizing, do not damage biological tissues, sensitive to weak resonances [2] such as the vibrational and rotational modes of many biomolecules [3], THz spectroscopy has soon become a promising technology for sensing research on proteins [4], virus [5] and antibiotics [6], DNA or RNA [7,8], and cells [9,10], etc. However, the detection sensitivity for recognizing a trace amount of biological samples is limited for most of the THz sensing systems. The main reason is that the wavelength of THz waves mismatches with the size of target biological molecules [11], resulting in a very weak electromagnetic response, and therefore leading to a poor scattering cross-section. Therefore, there is still a challenge to achieve efficiently THz specific detection of small biomolecules with high

sensitivity.

To solve this challenge, there are usually three approaches: (1) THz artificial microstructure, e.g. photonic crystals, plasmonics, and metasurfaces [12–18]. These artificial structures are developed to enhance the localized field, and then boost the interaction of sample and THz waves, thus improving the sensitivity. As a recent example, Zhang et al. proposed a biosensor consisting of THz cut wires and split-ring resonators, of which theoretical sensitivity can reach 496 GHz/RIU. Moreover, the mutant and wild-type glioma cells can be distinguished directly by this THz sensor [19]. (2) Introducing nanofilms, e.g. graphene, to improve sensitivity and achieve sensor tunability [20–23]. For example, Lee, et al. reported a graphene-assisted THz metamaterial for ssDNA sensing, and its detection accuracy reached nmol/mm² [20]. (3) Using gold nanoparticles (Au NPs) [24–26] to enhance the binding properties of the targeted biomolecules, so that the performance of sensors can be improved.

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Au NPs have been an important nanomaterial in biology and medicine because of their controllable geometry, optical and surface chemical properties [27–29]. Protein and Au NPs can form stable complexes by covalently binding without affecting protein activity and stability. It was also applied in THz immunosensing in recent years. For example, Ahmadivand *et al.* introduced Au NPs onto the toroidal THz plasmonic metasurface, which verified that the Au NPs can enhance capturing of biomarker molecules on the metallic metasurface, and up to 100-fold

sensitivity enhancement can be obtained in comparison to similar devices [30]. Then, the same scheme was also applied on a femtomolar-level detection of SARS-CoV-2 spike proteins [31]. This method provides exciting new ideas for trace sensing in the THz band and allows for the detection of low molecular-weight biomolecules, such as virus envelope protein and tag protein. In addition, it is worth noting that all-dielectric metasurfaces have been proved to have significant advantages in biosensing [32,33] due to richer electromagnetic modes,

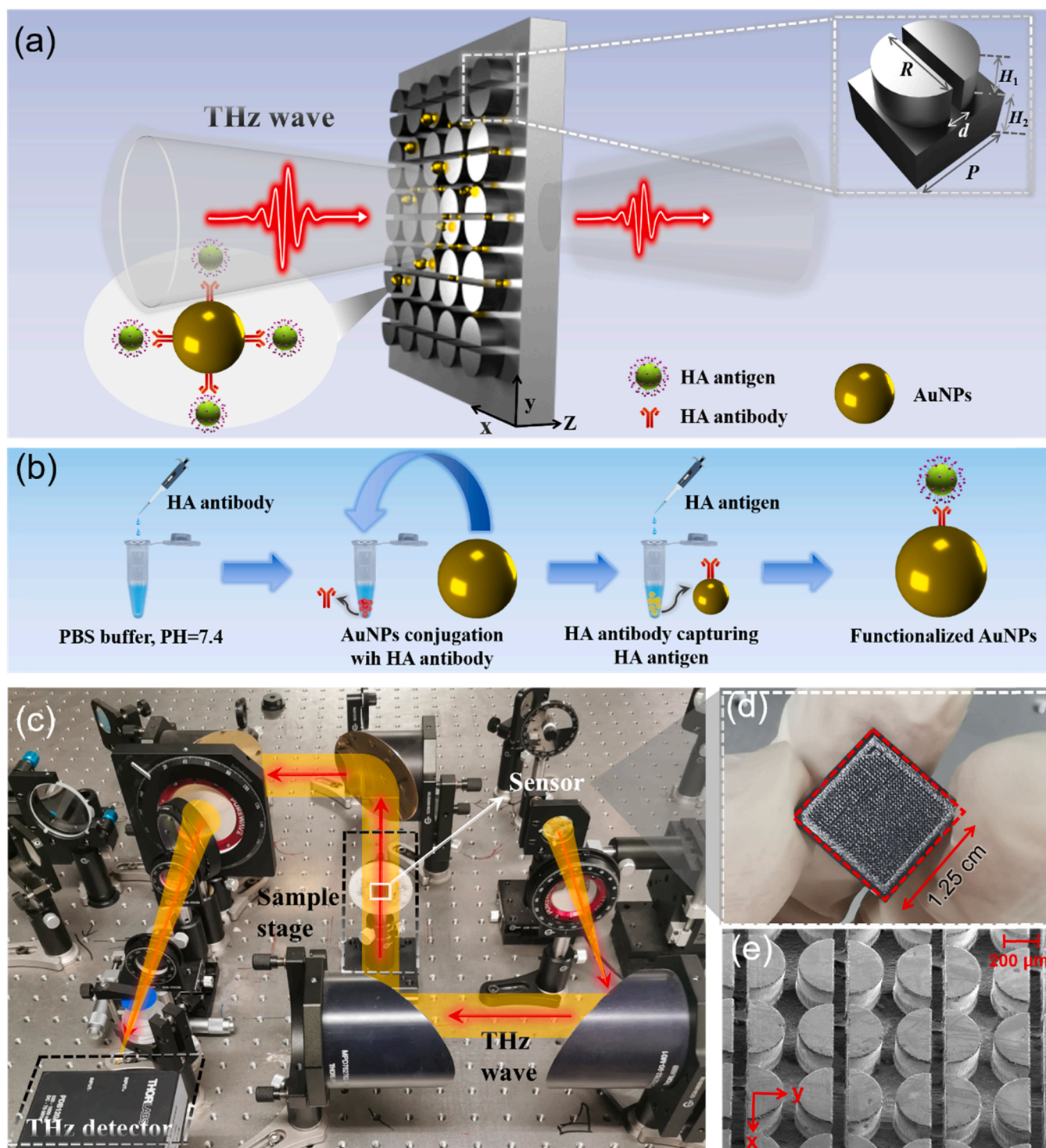


Fig. 1. (a) Schematic of the all-dielectric metasurface consists of split half-cylinders array, which is illuminated by a normally incident THz wave. (b) The schematic workflow of the functionalized Au NPs conjugated with the Anti-HA tag antibody. (c) THz-TDS system in our experiment. (d) The photo of the all-dielectric metasurface sensor chip with a size of $1.25\text{ cm} \times 1.25\text{ cm}$. (e) The SEM images of the all-dielectric metasurface.

higher Q value, and lower loss compared with the metallic metasurface. However, so far, there has been no experimental report on whether Au NPs assisted biochemical detection is effective for all-dielectric metasurface sensors.

Influenza hemagglutinin is a type of hemagglutinin found on the surface of influenza viruses. This type of hemagglutinin is a class of antigenic glycoproteins that enable the virus to bind to infected cells [34]. Detecting human influenza hemagglutinin tag protein (HA antigen) can help us screen the harmful influenza virus, which is key to disease diagnosis and prevention. Traditionally, fluorescence spectroscopy or electrochemical sensing are used for the identification of HA antigen [35]. Although THz sensing has achieved great success in biochemical measurement, there have been few reports on the immunosensing to HA antigen using THz technology so far.

In this work, we proposed a THz all-dielectric metasurface biosensor with the functionalized Au NPs to detect the HA antigen at nanomole-level concentration. All-dielectric metasurface provides a strong surface localized field enhancement, and it also increases the effective contact area of the sensor. These Au NPs are modified by the Anti-HA tag antibody to realize the specific binding of HA antigen. Two groups of contrast experiments have been performed: (1) by comparing the presence and absence of the functionalized Au NPs, the results show that this method can quantitatively detect HA antigen, and its sensitivity can be greatly improved by the functionalized Au NPs; (2) by comparing three other nonspecific proteins, the results show that only the HA antigen can be sensitively distinguished, which also proves the specificity of this method.

2. Methods

2.1. Design and fabrication of metasurface sensor

The geometry of the all-dielectric metasurface sensor is shown in Fig. 1a, of which the unit cell consists of a pair of split half-cylinders. Through the processes of masking, photolithography, and reactive ion beam etching on a high resistance silicon wafer, a deep etching relief pattern with a height of $H_1 = 250 \mu\text{m}$ was obtained. The period of the unit cell $P = 500 \mu\text{m}$, the separation distance between the two half-cylinders $d = 60 \mu\text{m}$. Each cylinder with a radius of $R = 200 \mu\text{m}$, the remaining substrate with a thickness of $H_2 = 250 \mu\text{m}$. The appearance of a single sensor chip is shown in Fig. 1d, with a size of $1.25 \text{ cm} \times 1.25 \text{ cm}$, and the SEM image of the metasurface is presented in Fig. 1e. This quasi three-dimensional structure is conducive to the attachment of biological samples and increases the effective surface area of the sensor.

2.2. Preparation of Au NPs with the Anti-HA tag antibody

According to our strategy shown in Fig. 1a, to improve the sensing performance, we introduced the Au NPs and dispersed them onto the surface of the sensor. The Au NPs can covalently conjugate to primary amines ($-\text{NH}_2$) of proteins [36], thus enhancing the detection of proteins, the Au NPs used in our experiments were produced by Sigma-Aldrich, we purchased it from J&K Scientific (Beijing, China). Fig. 1b shows the schematic workflow of the Au NPs conjugation with the Anti-HA tag antibody. By referring to the existing methods [30,31], and after several repeated tests, we summed up the optimized experimental steps and sample concentrations: First, for the conjugation of Anti-HA tag antibody (Cat. No. D191044, purchased from Sangon Biotech, Shanghai, China) with the Au NPs, we prepared reconstitution buffer by combining 1 ml of phosphate buffer solution (PBS, PH 7.4) with 50 μL of purified Anti-HA tag antibody, then added 1 ml of Au NPs solutions (with the particle average diameter of $\sim 80 \text{ nm}$). The mixed solution is centrifuged at 5500 rpm for about 5 min (low temperature, around 4°C). After centrifuging, the supernatant is removed and resuspended with 1 ml of PBS, right after the resuspended, the solution was sonicated for 2 min. By repeating the above centrifugation and sonication process at

least three times, the functionalized Au NPs conjugated with the Anti-HA tag antibody were successfully prepared. Second, a series of HA antigen solutions with concentrations ranging from 20 to 50 $\mu\text{g}/\text{ml}$ were prepared by dissolving an appropriate amount of HA antigen (Cat. No. D610004, purchased from Sangon Biotech, Shanghai, China) in PBS. At last, the final conjugated Au NPs samples were obtained by mixing the antigen solutions into the functionalized Au NPs samples.

In our experiment, 20 μL of the prepared Au NPs solution was dropped onto the surface of the sensor in each experiment, waiting for it to dry. And then we dropped the antigen solution with different concentrations onto the surface of the sensor. A thin film was formed on the surfaces of the sensor after the sample was dried. We measured the thickness of the film by a profilometer. As shown in Fig. 2, a groove is scratched in the film as a reference plane (green area in Fig. 2a), the measured thickness of protein film was about $2.5 \mu\text{m}$, as can be seen from Fig. 2b. Additionally, the SEM images of protein film and functionalized Au NPs in the gap of the sensor are shown in Fig. 2c and Fig. 2d, respectively. It's important to emphasize that the total amount of HA antigen on the metasurface is indeed independent of the presence or absence of AuNPs, but its different distribution location can cause different sensing results. If there are no AuNPs, this film can be considered as a uniform HA antigen protein film. When there are AuNPs in the sample solution, during the drying process of the solution, the AuNPs will sink to the bottom of the solution, the HA antigens bound to it will also sink with it, in this case, more HA antigens are accumulated in the gap and bottom of the sensor.

2.3. THz experimental system

The sensing experiments in this work were performed by a THz time-domain spectroscopy (THz-TDS) system at room temperature of about 25°C and a humidity of 10%. Fig. 1c shows the photo of the system. A femtosecond laser with a wavelength of 800 nm is incident on a GaAs photoconductive antenna, which emits a linearly polarized THz pulse. The pulse is then focused on the sensor. By varying the time delay between the THz waves and the probe laser beam, both amplitude and phase information can be measured. Finally, the frequency spectrum $A_s(\omega)$ can be obtained by applying Fourier transform to the time domain signal $E_s(t)$. The intensity spectra of the detected samples can be calculated by $T(\omega) = 20 \log A_s(\omega)/A_{\text{ref}}(\omega)$, where $A_s(\omega)$ is the amplitude transmission spectrum after THz wave passing through the sample, and A_{ref} represents the amplitude of air without any sample as a reference. The time resolution of the measured THz pulse is 0.04 ps, and the effective working frequency range of our system was 0.05–3 THz, with a detection signal-to-noise (S/N) ratio of over $10^4:1$.

3. Results and discussion

3.1. Simulation analysis

The numerical simulations of the sensor were performed by using the time-domain solver of the commercial software CST Microwave Studio. The geometry of the metasurface unit was modeled according to Fig. 1a, a pair of periodic boundary conditions were set in this simulation model, and the dielectric permittivity of silicon was set as $\epsilon = 11.8$. The electromagnetic responses of the metasurface in both x and y polarization directions were simulated as shown in Fig. 3. As shown in Fig. 3b, there are two resonances at 0.42 THz and 0.92 THz for y-polarization, and also two resonances at 0.40 THz and 0.93 THz for x-polarization. To distinguish these two resonance modes conveniently, the resonance mode at the lower frequency is marked f_1 and the higher one is f_2 .

The electric field and magnetic field distributions of the two resonance modes were monitored as shown in Figs. 3c–3f. For the x-polarization shown in Figs. 3c and 3e, at the resonance f_1 , there is no obvious localized electric field inside the dielectric cylinders, especially in the gap between the two half-cylinders; at the resonance f_2 , in the center of

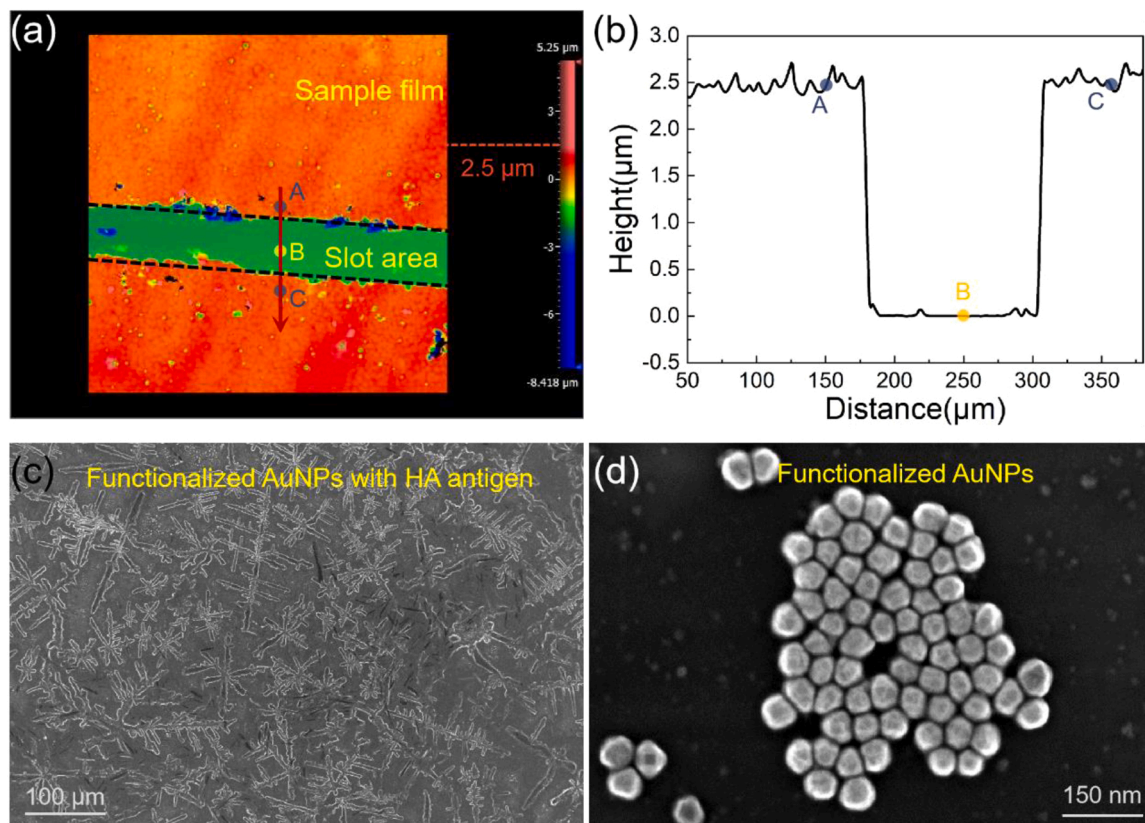


Fig. 2. (a) The pattern of protein film on the metasurface measured by the profilometer, and (b) the corresponding thickness of the film relative to the surface of the sensor at the position of the red line in Fig. 2a. SEM images of (c) the HA antigen film and (d) the functionalized Au NPs, respectively. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

the unit cell, the direction of the induced electric field vectors is opposite to the incident electric field, and generate a higher field enhancement in the gap between the two half-cylinders compared to that of f_1 , which indicates that the resonance f_2 will have higher sensitivity than the resonance f_1 . It can be also clearly seen that the magnetic field is almost confined inside the structure. Most of the electromagnetic field is localized in the dielectric cylinders or not in the gap between the two half-cylinders, which is not conducive to sensing.

For y -polarization shown in Fig. 3d, we find that toroidal electric fields appear at both resonance f_1 and resonance f_2 . The formation of these head-to-tail electric vectors generates the toroidal dipole-like response, so most of the electric field energy is confined to the gap between the two half-cylinders. Further, a significant magnetic field enhancement is also observed in the same area in Fig. 3f, thus the sensitivity of the sensor in the y -polarized case is better than the x -polarization. Additionally, at the resonance f_2 , the electromagnetic field in the gap is more concentrated than the resonance f_1 , which makes the resonance f_2 more sensitive to the changes of the dielectric environment around the structure.

Then we simulated the transmission spectra of the sensor filled with the simulative medium of different refractive indexes as shown in Fig. 4. We set the simulative medium filled in the grooves and tops of the metasurface with the thickness of 3 μm , and the simulative medium was considered to be loss-free. With the increase of the refractive index from 1.5 to 1.8, both f_1 and f_2 move towards higher frequencies. The frequency shift of resonance f_2 is much larger than that of the resonance f_1 , because the electric field distribution of the resonance f_2 in the grooves is more concentrated than that of the resonance f_1 in both y - and x -polarization incident wave. Moreover, in Fig. 4a, the resonance f_2 varies with the refractive index from 0.92 THz to 1.05 THz, but in Fig. 4b, it varies from 0.93 THz to 1.02 THz. Through the calculation, the

sensitivity of the sensor is 112.5 GHz/RIU for the x -polarization, and 162.5 GHz/RIU for the y -polarization, which demonstrates the sensitivity in the y -polarization is higher 30% than the x -polarized case. This result is consistent with the conclusion of the previous mode analysis.

3.2. Experimental sensing of HA antigen protein with different concentrations

The experimental transmission spectra of the blank metasurface sensor are also shown in Fig. 3b, and we can find the experimental results are well confirmed by the simulation results. The slight difference comes from material loss and fabrication error. The maximum error between the experiment and simulation is 15 GHz for f_2 in x -polarization, which is mainly caused by the error of the etching depth. Compared with the simulation results, the position of resonance appears a slight shift, but the mechanism of resonance does not change, it still conforms to our simulation prediction, so it does not affect the analysis for the actual sensitivity of x -polarization. Next, we performed the THz sensing experiments for the HA antigen protein with different concentrations. In our experiment, Au NPs were modified with the Anti-HA tag antibody to form the functionalized Au NPs for specifically recognizing the target HA antigen (see Section 2.2 for the detailed sample preparation). In the following text and figures, the Au NPs specifically refer to the functionalized Au NPs modified with the Anti-HA tag antibody. We have also pointed out that there is a strong localized field enhancement in the gap between two half-cylinders, and it can be noticed from Fig. 2d that the Au NPs form the clusters in the gap, which further improves the electromagnetic response of the sensor to the biomolecules.

Detecting HA antigen with different concentrations from 20 to 50 $\mu\text{g/ml}$ for the y -polarization is presented in Fig. 5. To evaluate the effects of the functionalized Au NPs, we also detected the HA antigen

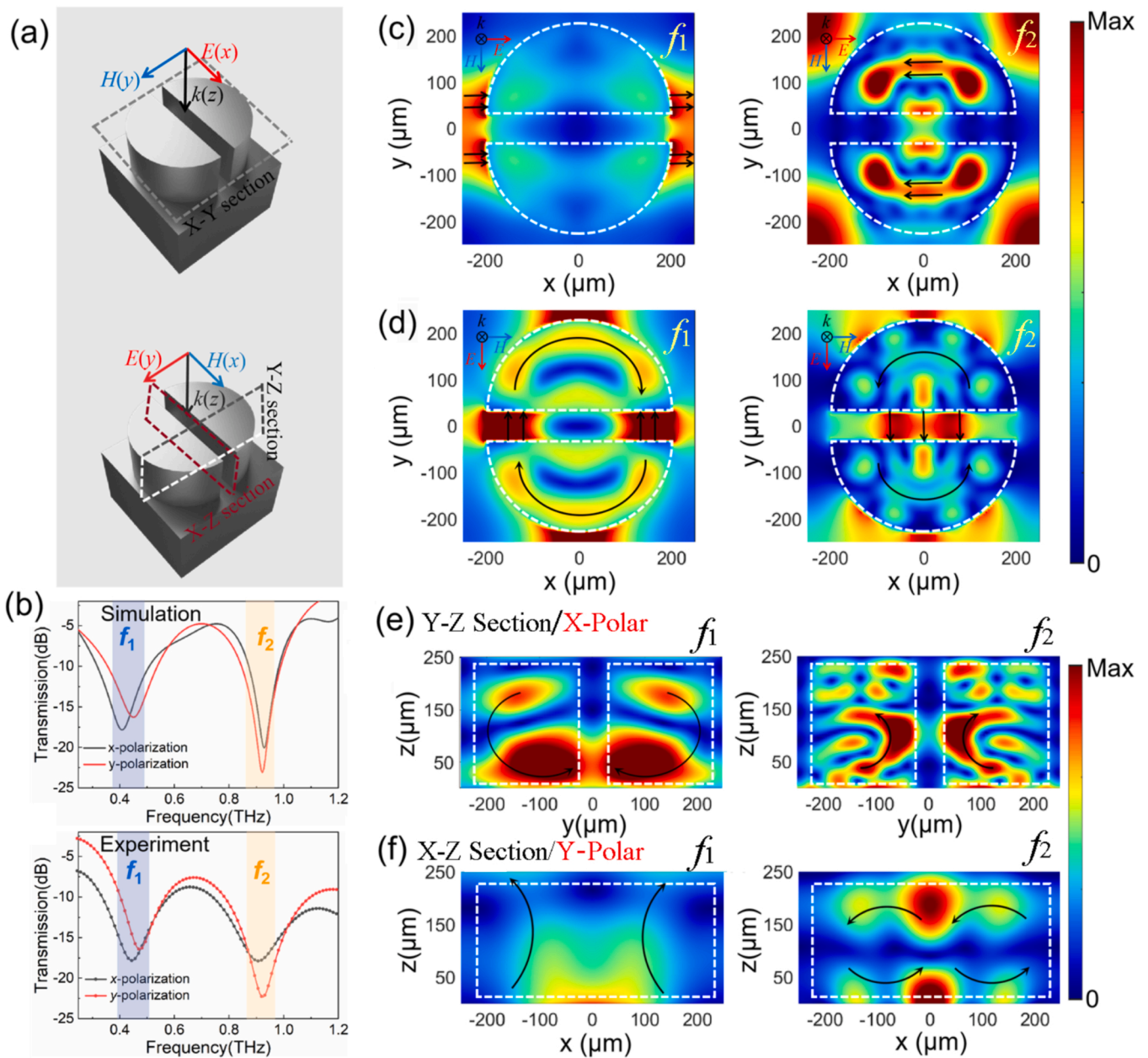


Fig. 3. (a) Schematic of a unit cell of the metasurface-based sensor. The dotted box in the figure represents the section plane where the electromagnetic field is recorded. (b) Simulated and experimental transmission spectra for the blank metasurface sensor for x and y polarizations, respectively. Electric field distributions in the X-Y section at f_1 and f_2 for (c) x- and (d) y-polarizations. (e) Magnetic field distributions in the Y-Z section at f_1 and f_2 for x-polarization. (f) Magnetic field distributions in the X-Z section at f_1 and f_2 for y-polarization.

without Au NPs as a comparison. From Fig. 5, we observed an obvious frequency shift of the resonances with the increase of the HA antigen concentrations. In the presence of the Au NPs, the resonance f_1 has almost no frequency shift, but the resonance f_2 shifted to 0.95 THz because the formation of the Au NPs layer on the surface of the sensor affects the surrounding dielectric environment and the resonance shifting. This is consistent with our previous theoretical analysis: the resonance f_2 has a stronger response to the changes of the surrounding dielectric environment due to more concentrated electromagnetic field and toroidal dipole-like response. Thus, the impact on the f_2 is greater than the impact on the f_1 . For the solution composed of the functionalized Au NPs and HA antigen with a concentration of 50 $\mu\text{g/ml}$, the resonance f_1 shifted to 0.513 THz, and the resonance f_2 can be moved to 1.037 THz. Compared Fig. 5a with Fig. 5b, it's clear that the sensing results based on functionalized Au NPs have a more remarkable change

in the frequency shift, which is clearly distinguished between different HA antigen concentrations. Thus, we can conclude that the presence of the functionalized Au NPs conjugated with Anti-HA tag antibody helps improve the HA antigen protein capture effectively, and the sensing performance has also been greatly improved.

Similar results of detecting HA antigen with different concentrations for x-polarization are presented in Fig. 6: the resonance f_2 shifted from 0.90 THz to 0.975 THz with Au NPs and 0.931 THz without Au NPs, respectively. Thus, with the assistance of the functionalized Au NPs, the sensitivity of the sensor is also improved effectively for the x-polarization. But compared with the y-polarized case, the value of the frequency shift is reduced. According to Figs. 5c, 5d, 6c, and 6d, we can compare the difference of frequency shift with concentration for two polarization directions: the total frequency shifts of f_2 in the y-polarized case is 87.5 GHz, about 1.56 times than 56.25 GHz in x-polarized one.

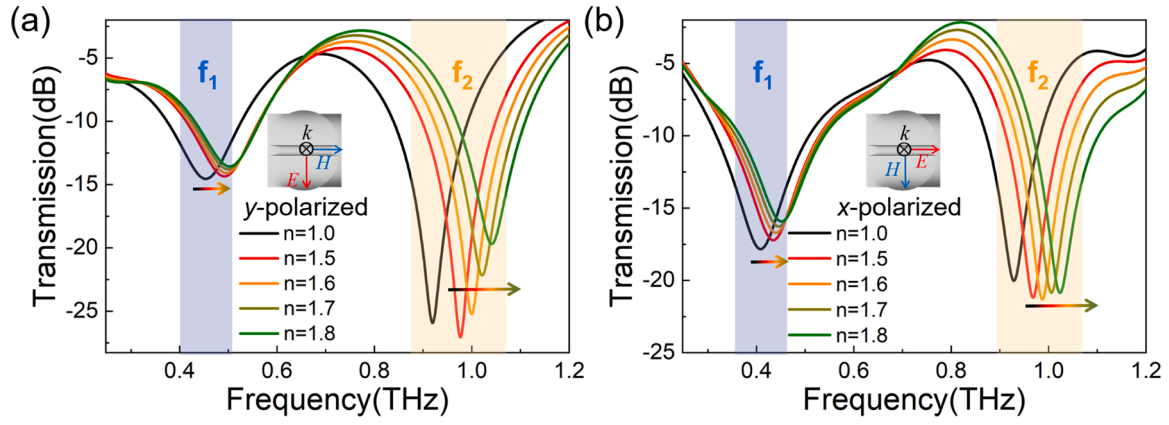


Fig. 4. Simulative transmission spectra of the blank sensor (black line) and filled with the medium of different refractive indexes from 1.5 to 1.8. The polarization direction of the incident waves is along the (a) y- and (b) x-axis.

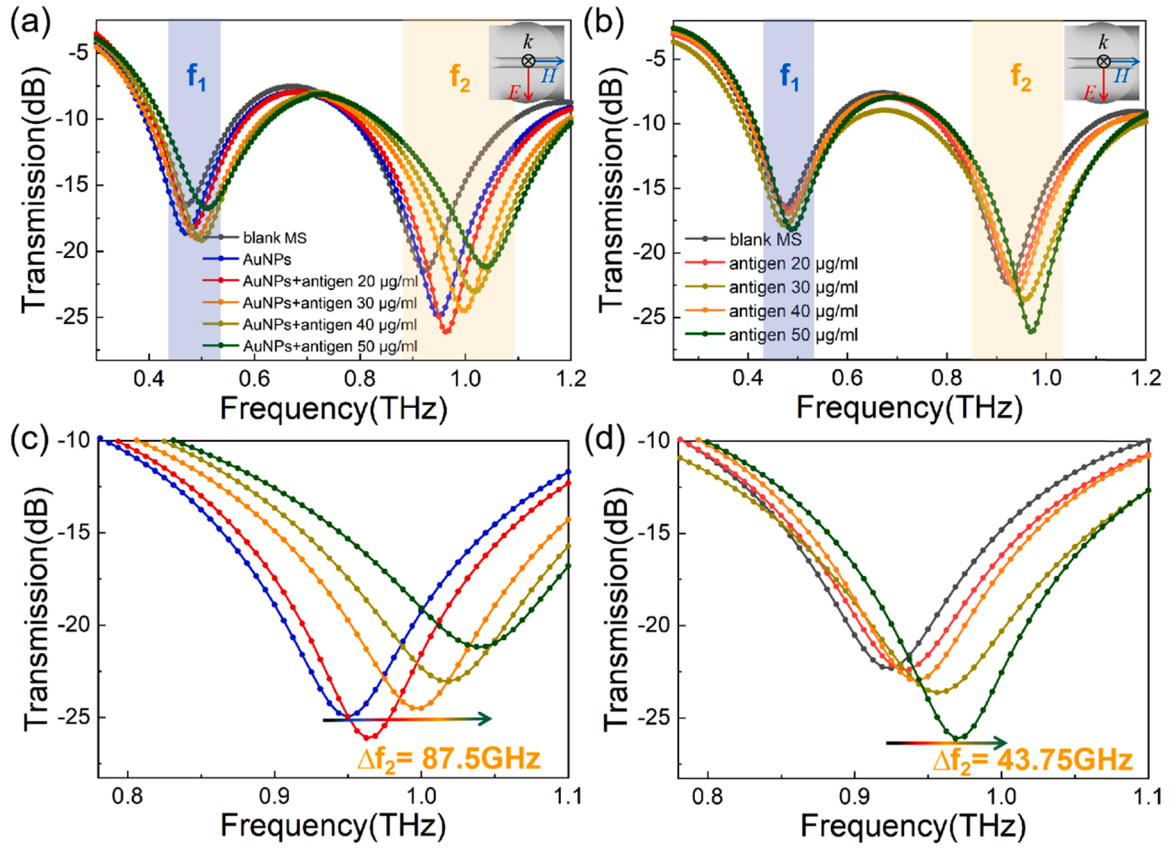


Fig. 5. Experimental transmission spectra of y-polarization for HA antigen from 20 $\mu\text{g/ml}$ to 50 $\mu\text{g/ml}$ on metasurface sensors (a) with and (b) without the functionalized Au NPs. The enlarged view of the frequency shifts for the resonance f_2 (c) with and (d) without the functionalized Au NPs.

To better compare the sensing performance of the sensor for two different polarization directions, we plotted the resonance frequency v.s concentration and its fitting line in Fig. 7. The resonance frequency shows a good linear shift trend with the increase of concentration, indicating that our sensor has stable and reliable sensing performance. The slope of the fitting curve expresses the sensitivity of the device, and the sensitivity S_{f1} and S_{f2} are defined as:

$$S_f = \frac{\partial f}{\partial M} \quad (1)$$

where f is the frequency of the resonance, M represents the Molar mass of HA antigen filled onto the sensor, the detailed results are shown in

Table 1. Note that the molecular weight of HA antigen is ~ 1.11 kDa [37].

As shown in Table 1, for the y-polarized case, higher values of S_{f1} and S_{f2} are obtained when the functionalized Au NPs are present in the sample. After the antigen is captured by the functionalized Au NPs, the sensitivity of the f_1 increases from 0.46 GHz·ml/nmol to 0.83 GHz·ml/nmol; the sensitivity of the f_2 increases from 1.11 to 2.96 GHz·ml/nmol, with 2.66 times than that of without Au NPs. For the x-polarized case, the sensitivity varied from 0.97 to 1.85 GHz·ml/nmol, with 1.91 times. The minimum frequency step size M_f of the THz-TDS system is 3.125 GHz, so the detection accuracy based on our system can be calculated as follows.

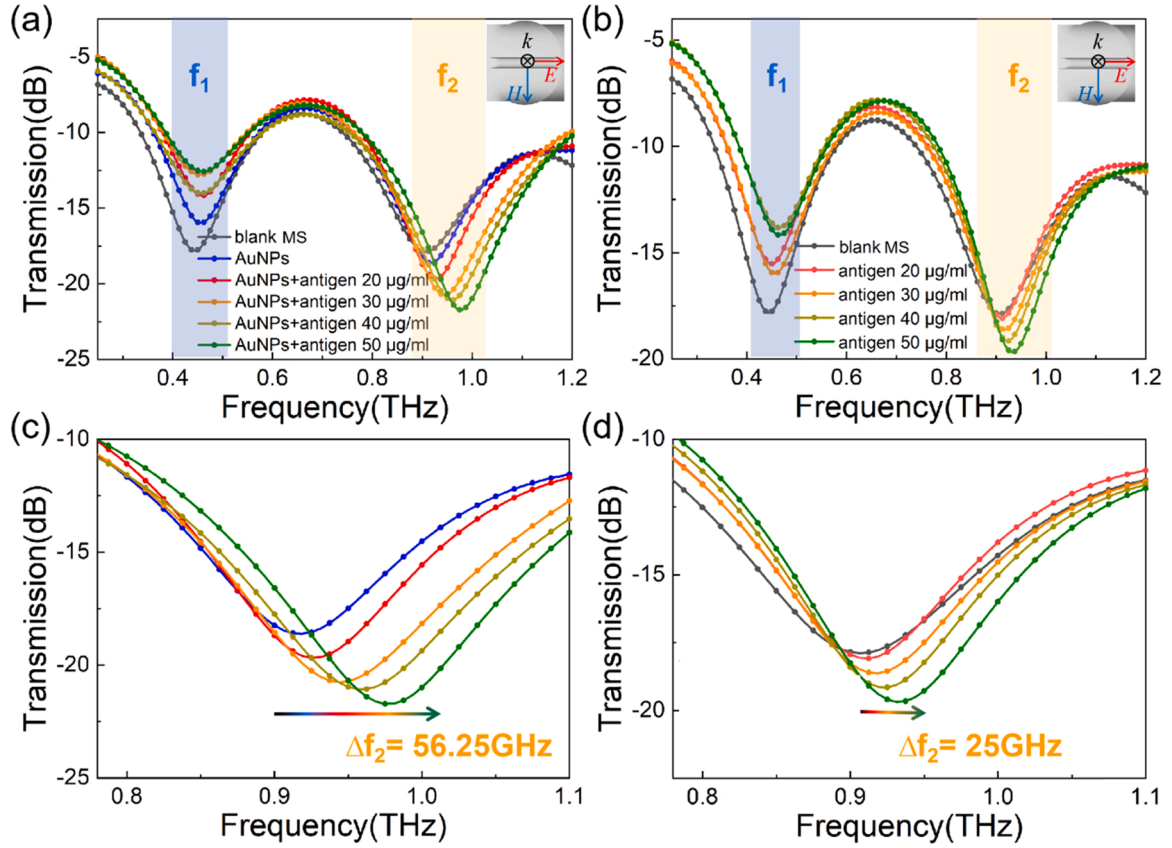


Fig. 6. Experimental transmission spectra of x-polarization for HA antigen from 20 $\mu\text{g/ml}$ to 50 $\mu\text{g/ml}$ on metasurface sensors (a) with and (b) without the functionalized Au NPs. (c) The enlarged view of the frequency shifts for the resonance f_2 with the functionalized Au NPs and (d) without the functionalized Au NPs.

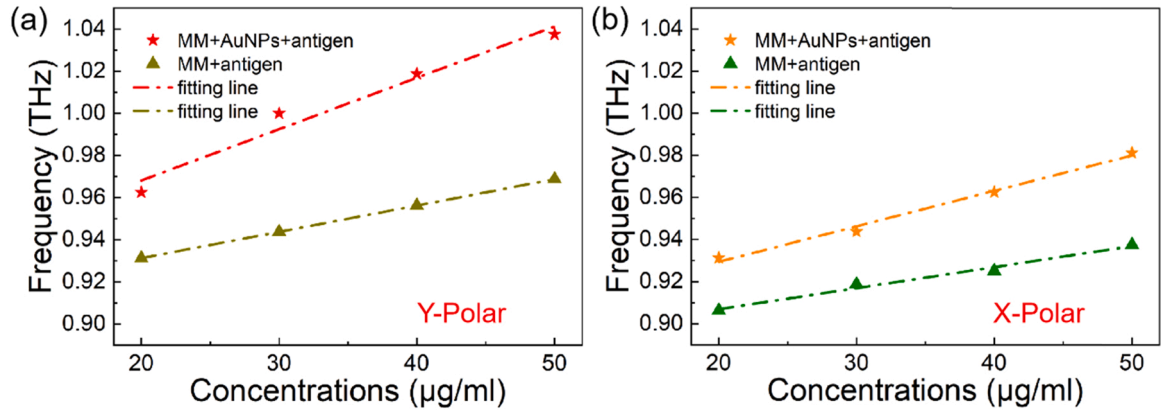


Fig. 7. Frequency shifts of f_2 with the change of sample concentrations, (a) for y-polarization and (b) x-polarization. Dotted lines are linear fits to the data.

Table 1

The sensing performance for HA antigen with and without Au NPs.

Polar	Au NPs	S_{f1} (GHz•ml/nmol)	M_{c1} (nmol/ml)	S_{f2} (GHz•ml/nmol)	M_{c2} (nmol/ml)
Y	W/O	0.46	6.79	1.11	2.81
Y	W	0.83	3.76	2.96	1.05
X	W/O	–	–	0.97	3.22
X	W	–	–	1.86	1.68

$$M_c = \frac{1}{FOM} = \frac{M_f}{S_f} \quad (2)$$

where FOM (Figure of merit) is the quality factor of the sample in the sensing system, and M_c is detection accuracy. According to Eq. 2, the highest detection accuracy reaches 1.05 nmol/ml with the presence of Au NPs in the y-polarization and 1.68 nmol/ml in the x-polarized case. Compared with the y-polarized case, the sensitivity and detection accuracy are reduced. These experimental results are consistent with the simulation results in Section 3.1, which also confirms the influence of the electromagnetic distribution of the metasurface sensor on the sensing performance. All the results indicate the sensing performance is

improved with the assistance of the functionalized Au NPs, and also show the sensing performance of y-polarization is better than that of the x-polarization. Thus, with the assistance of AuNPs, we have realized the detection of HA-tag protein with the molecular weight is only ~ 1.11 kDa, which is much smaller than other proteins samples with molecular weights in the tens or even hundreds kDa, this is the main advantage of our sensing strategy.

3.3. Specificity verification by comparison with other proteins

For biosensing applications, it is especially crucial to evaluate the selective detection capability of the sensor for target biomolecules, this is of great significance to biological diagnosis and treatment, disease detection, and other practical applications. To verify the specific recognition and selective detection ability of our sensing method for HA antigen, we experimentally measured other three proteins: bovine serum protein (BSA), ovalbumin (OVA), and whey protein (WP) with the same concentration of $50 \mu\text{g/ml}$ on the metasurface sensor modified by the functionalized Au NPs conjugated with the Anti-HA tag antibody, these three proteins do not bind specifically to HA antibodies, therefore, specific recognition can be verified by comparing them with HA antigen. In each experiment, we dropped the same volumes and concentration of various proteins on the sensor, and the sensor was cleaned 3–4 times before measuring different kinds of protein, so the experimental results would not be affected by other unbound proteins. The experiments results were shown in Fig. 8. The frequency shift for HA antigen is much larger than that of the other three proteins. Compared to the blank metasurface sensor with the Au NPs, the spectral changes of the other three proteins are very small. They nearly overlap and cannot be distinguished among them. In other words, this metasurface sensor has a highly sensitive electromagnetic response only to HA antigen. The detailed values of the frequency shift for the four different proteins are summarized in Table 2.

For the y-polarization, the frequency shift for HA antigen is 87.5 GHz, for other proteins are lower than 25 GHz; For the x-polarization, the frequency shift for HA antigen is 56.2 GHz, for other proteins are below 20 GHz. Such a large difference in the frequency shift derives from the specific recognition and binding of HA antigen by functionalized Au NPs. The functionalized Au NPs conjugated with Anti-HA tag antibody have been modified on the surface of the sensor before the measurement. When the HA antigen was dropped onto the sensor surface, the HA antigen molecules can bind closely to the Au NPs due to the protein immune reaction, thus greatly improving the sensor efficiency. In other words, because other proteins do not react with the Anti-HA tag antibody, they cannot be bound by the functionalized Au NPs. Although their concentration is the same as the HA antigen, only a few protein molecules can contact the field-enhanced region on the sensor surface,

Table 2

The frequency shift of the sensor for the four proteins.

Polarization-direction	BSA	OVA	WP	HA antigen
$\Delta f_{y\text{-polar}}$ (GHz)	25	19	18	87.5
$\Delta f_{x\text{-polar}}$ (GHz)	19	12	12	56.2

especially in the gap of the sensor. Therefore, the near-field dielectric environment of the sensor surface does not change significantly, which leads to being insensitive to the response of these proteins. To sum up, according to our experimental scheme, the sensor shows specific detection of the target biomolecule and can distinguish the target biomolecule from other protein molecules.

4. Conclusion

In conclusion, a THz immunospecific sensing for HA antigen has been performed on an all-dielectric metasurface with the functionalized Au NPs. Simulation analysis shows that the metasurface sensor with the split half-cylinders array has a strong surface field localization effect at a certain polarization and frequency. The introduction of Au NPs effectively immobilized the Anti-HA tag antibody on the surface of the metasurface, and further enhanced the field localization and interaction with the measured object. By the contrast experiments, the sensitivity of the sensor has been improved significantly to the HA antigen, 2.66 times higher than without the functionalized Au NPs. The maximum sensitivity for HA antigen concentration reaches $2.96 \text{ GHz}\cdot\text{ml/nmol}$, and the detection accuracy is up to 1.05 nmol/ml . Furthermore, by comparing the sensing performance for the BSA, OVA, and WP proteins with the same concentration, the results show that only HA antigen can be distinguished significantly, and its detection sensitivity is much higher than the other three proteins, so the specificity of this method has been confirmed. Therefore, the proposed sensing platform based on the all-dielectric metasurface and the functionalized Au NPs has been proved to be effective for THz immunospecific sensing, which is expected to provide new methods and ideas for the application of biochemical detection.

CRediT authorship contribution statement

Weinan Shi: Investigation, Methodology, Data curation, Writing – original draft. **Fei Fan:** Conceptualization, Methodology, Validation, Writing – review & editing, Project administration. **Shanshan Li:** Resources. **Ziyang Zhang:** Investigation, Data curation. **Hongliang Liu:** Resources. **Xianghui Wang:** Conceptualization, Methodology, Project administration. **Shengjiang Chang:** Supervision, Project administration.

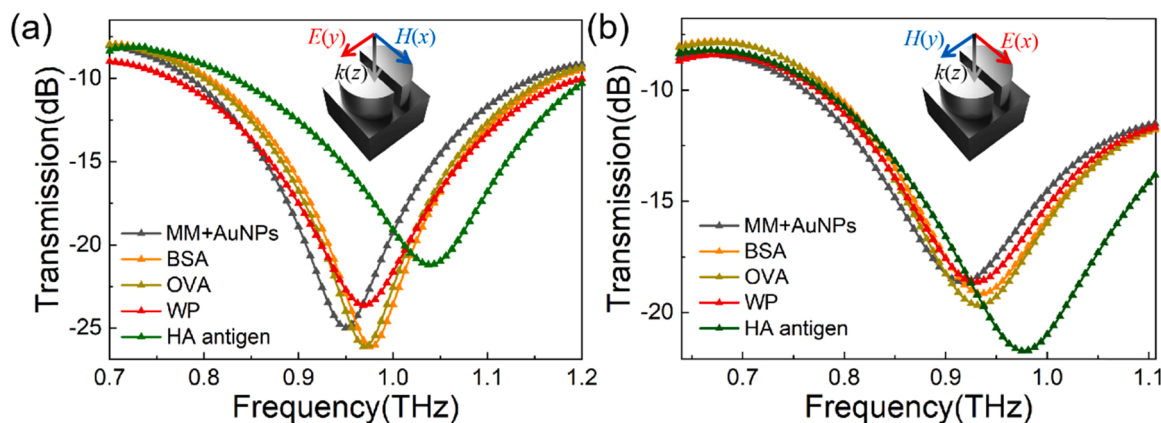


Fig. 8. Experimental transmission spectra for the $50 \mu\text{g/ml}$ BSA, OVA, WP, and HA antigen on the metasurface sensor all modified with the functionalized Au NPs. The black line is the blank metasurface sensor with the functionalized Au NPs without protein samples. (a) y-polarization; (b) x-polarization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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